Abstract:

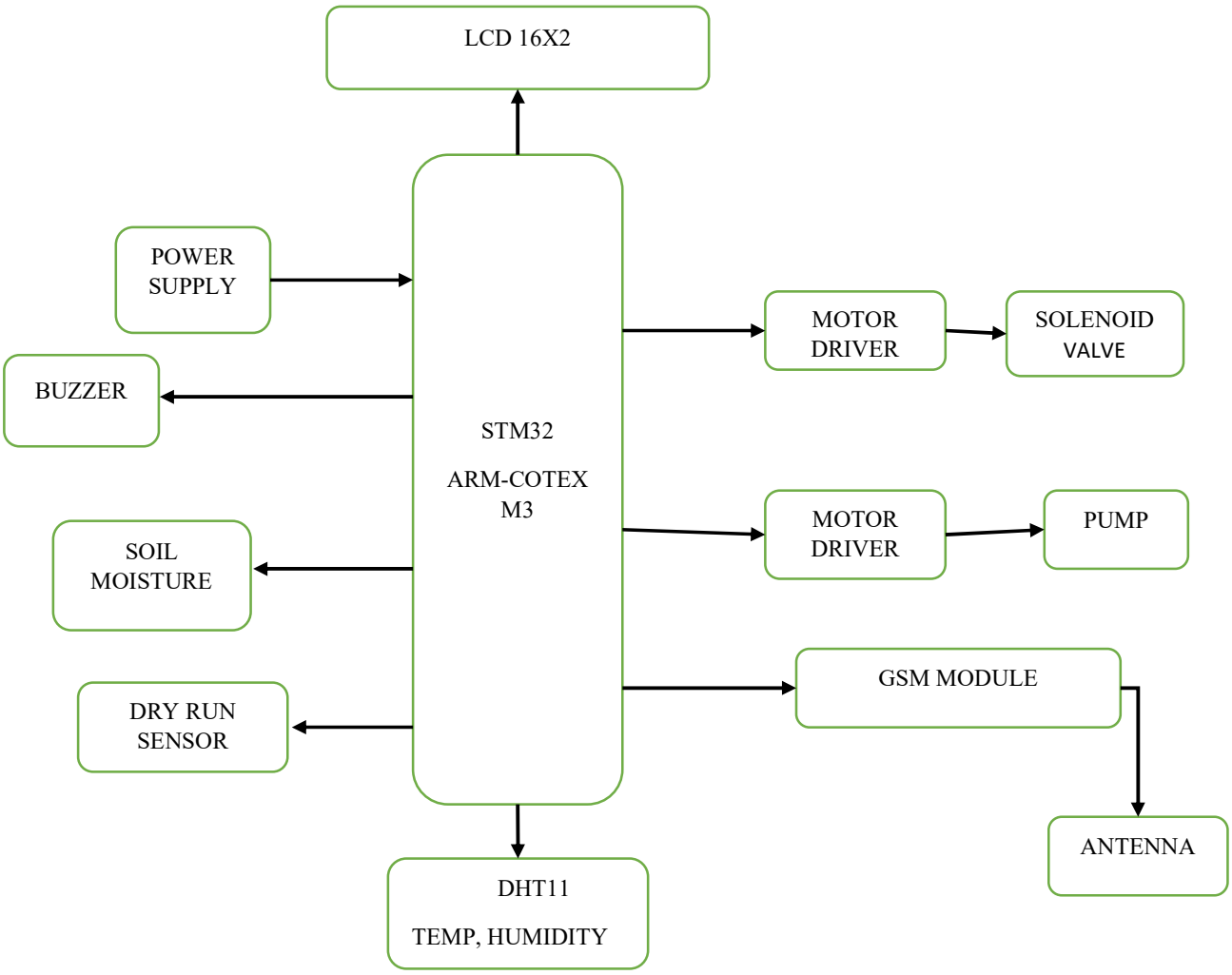
This GSM-based IoT irrigation system uses an STM32 microcontroller and SIM800L module to enable remote control via SMS. Farmers can start irrigation by specifying the field and duration. A relay operates the water pump, and solenoid valves direct water flow. Soil moisture sensors stop irrigation when optimal levels are reached, while a dry run sensor protects the pump. A timer ensures auto shut-off, and SMS updates are sent after each session. The system offers a cost-effective, efficient, and sustainable solution for multi-field farms.

1.Introduction:

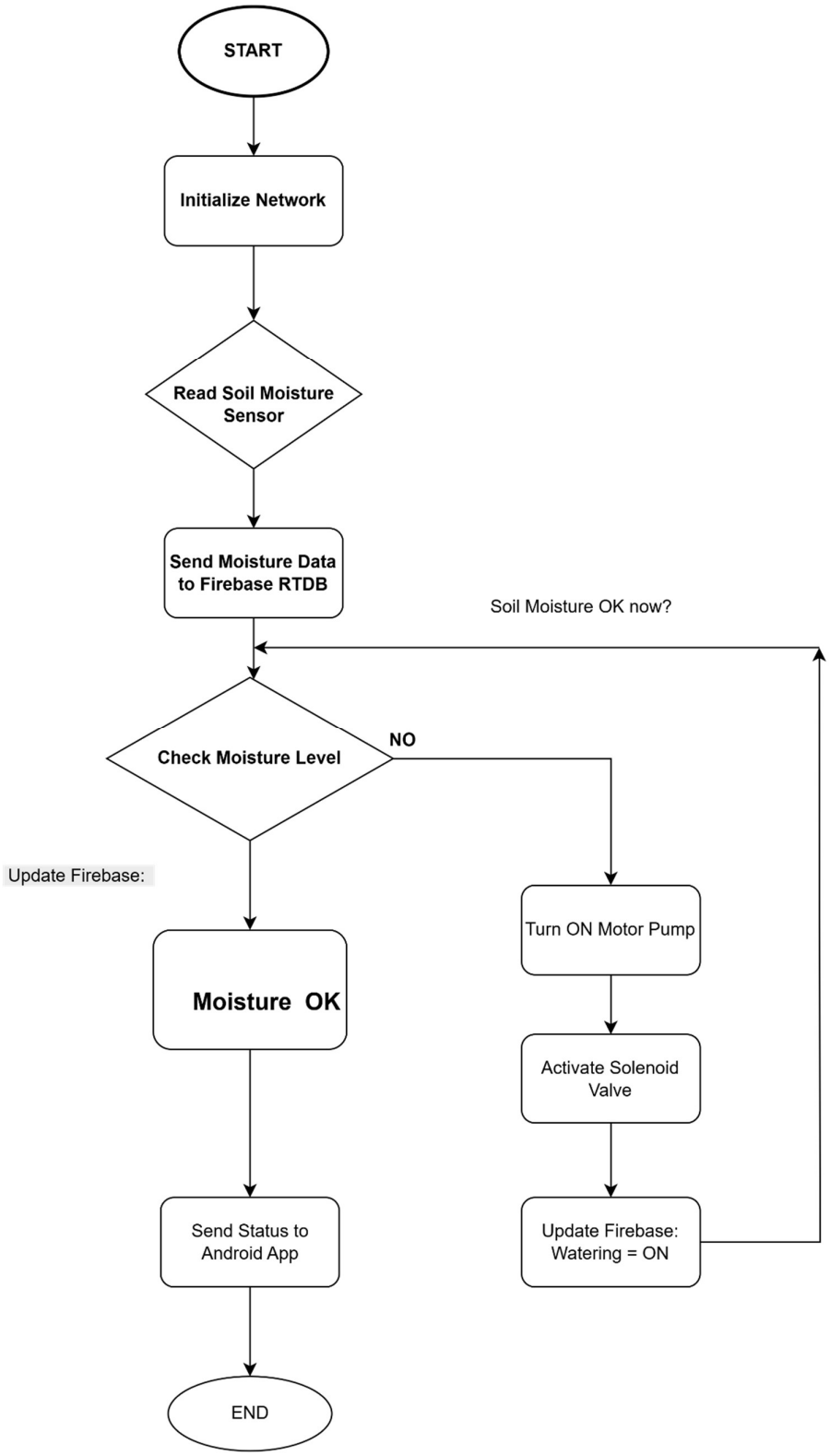
Agriculture, especially in countries like India, faces challenges like water wastage, labor dependence, and inefficient irrigation. IoT-based smart irrigation offers a solution by automating water distribution based on crop needs. These systems can be remotely controlled via SMS or mobile apps, allowing farmers to manage multiple fields efficiently. Key features include dry run protection to prevent pump damage, and timer-based irrigation to avoid overwatering. This technology enhances water use, reduces manual effort, and supports sustainable farming.

smart irrigation systems offer a sustainable solution by ensuring that water is used only when and where it is needed. This project introduces an IoT-enabled smart irrigation system that can be controlled via SMS, making it suitable for areas with low internet availability. The system is designed to manage multiple fields individually, allowing farmers to irrigate different zones based on soil moisture conditions and crop requirements.

2. Block Diagram:



3.Flow chart:


START
Initialize Network
Read Soil Moisture Sensor
Send Moisture Data to Firebase RTDB
Soil Moisture OK now?
Check Moisture Level
NO
Update Firebase:
Moisture OK
Turn ON Motor Pump
Activate Solenoid Valve
Send Status to Android App
Update Firebase: Watering = ON
END